

Figure 1 | Each fits with a partner—but not with each other

One locus shown • complete observed allele set: {1, 2, 3, 4}

Exactly two diploid contributors; complete allele observation; no dropout, drop-in, or error.

Configuration Two allele copies per person One-locus LR

A + unknown All four alleles explained	A 2 3 + Required partner 1 4	11.5741
B + unknown All four alleles explained	B 1 3 + Required partner 2 4	13.2275
A + B Observed allele 4 unexplained	A 2 3 + B 1 3	0

Symbols represent allele copies—not peak heights. Each separate comparison uses a different unknown partner.

Full profile: thirteen independently specified loci

A + unknown: $LR \approx 6.69 \times 10^{13}$

B + unknown: $LR \approx 3.80 \times 10^{14}$

A + B: $LR = 0$

LRs compare each stated configuration against two unknown contributors; these are selected comparisons.

Unknown identities exclude A and B, not their genotypes. Contributor positions are exchangeable labels.

The joint zero concerns the named pair as the complete configuration under this strict two-person model.

It does not exclude contribution somewhere within a larger mixture or under a different observation model.

Hardy–Weinberg allele frequencies for 1–5: 0.07, 0.08, 0.09, 0.10, 0.66.

Full-profile values refer to thirteen loci, not just the single locus illustrated above.

Figure 2 | Three population filters

Different genotype sets—not three estimates of one probability

Observed allele set: {1, 2, 3, 4}



Blue: target genotype; $f(t)$ denotes its population frequency. Green: other compatible heterozygotes. Gray: homozygotes.

Cell areas do not represent population probabilities. Outside-set genotypes are not displayed.

None of these quantities is a posterior probability of contribution.

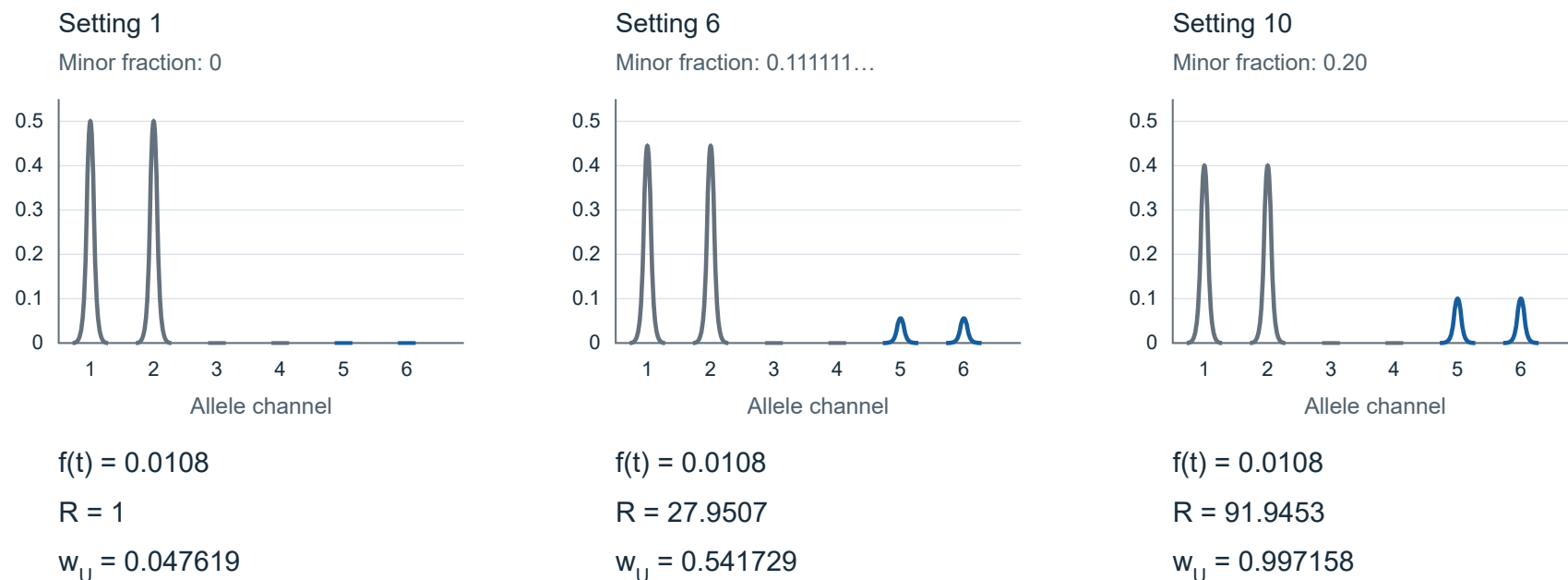
Strict model: exactly two diploid contributors; complete allele observation; no dropout, drop-in, or error.

Hardy–Weinberg population genotypes. Allele frequencies for 1–5: 0.07, 0.08, 0.09, 0.10, 0.66.

Figure 3 | Fixed genotype rarity, changing minor signal

Known major genotype: 1/2 • target in minor position: 5/6

Synthetic signal (arbitrary units) • common axes • $f(t)$: target genotype frequency



Blue: target allele channels 5 and 6. Peak heights are the supplied synthetic signals.

w_U : uniform target-genotype prior; not a posterior probability of named-person contribution.

Fractions are prescribed inputs. Zero minor fraction is an uninformative boundary—not dropout or exclusion.

Independent Gaussian channel errors; $\sigma = 0.035$. No detection threshold is applied.

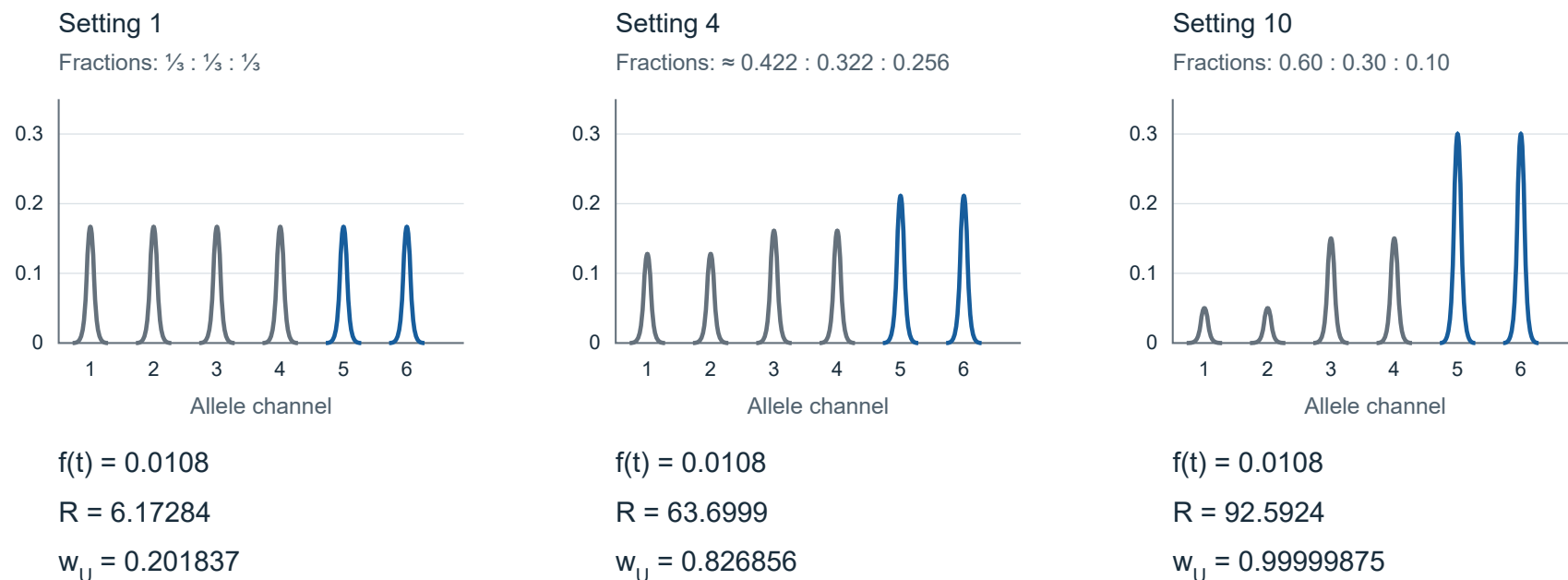
Same toy locus throughout—not three independent loci. Displayed statistics are rounded.

Schematic peak shapes; widths are illustrative and are not part of the statistical model.

Figure 4 | The same six alleles, different quantitative patterns

Target proposition fixes 5/6 in the first contributor position

Synthetic signal (arbitrary units) • common axes • $f(t)$: target genotype frequency



Generator order: 5/6, 3/4, 1/2. Generator truth is not treated as known during evaluation.

Other contributor genotypes are population-weighted; fractions are fixed inputs under both propositions.

w_U : uniform target-genotype prior; not a posterior probability of named-person contribution.

Blue: channels 5 and 6. All six channels have positive signal in every setting.

Independent Gaussian channel errors; $\sigma = 0.035$. Same toy locus—not independent loci.

Peak heights use supplied arrays; displayed fractions and statistics are rounded.

Schematic peak shapes; widths are illustrative and are not part of the statistical model.